Qualifying Paper: Large-scale causal discovery using interventional data sheds light on gene network structure in K562 cells

Extending the sparse direct causal-effect network to reconstruct total causal effects with greater estimation accuracy and better structural recovery

by

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1 Problem Definition

Causal discovery among gene regulatory pathways is a challenging problem for researchers due to the high dimensionality of the regulatory networks. Conducting randomized experiments to investigate every possible gene-gene relationship is impractical [1]. However, relying solely on observational data presents challenges due to measured and unmeasured confounding, reverse causation, and complex regulatory structures such as feedback and feedforward loops, so the exact underlying network cannot generally be identified [1]. Searching through all possible causal graphs is computationally intractable, as the number of possible network structures grows exponentially with the number of variables [2].

Interventional CRISPR perturbation data can be used to mitigate the limitations in causal identifiability [3]. In this context, an intervention is realized by disrupting a targeted gene using a guide RNA (i.e., through a knock-out), which alters the gene's original causal dependencies [2]. The resulting transcriptional responses are then used to estimate the average causal effects (ACEs) between genes [3]. These experiments produce two input matrices: Y, an $N \times D$ matrix of gene expression measurements, and X, a binary $N \times Q$ matrix indicating which genes are targeted by guide RNAs in each cell. N denotes the number of cells, D denotes the number of genes, and Q denotes the number of guide RNA targets.

The overarching statistical problem that Brown et al. [3] seek to investigate is **the estimation of direct causal effects between genes using interventional data**. This paper specifies a linear structural equation model,

$$Y = YG + X\beta + \gamma, \tag{1}$$

where G is a $D \times D$ matrix of direct causal effects between genes, β is a $Q \times D$ matrix representing the effects of guide RNA interventions, and γ is an $N \times D$ matrix of random noise. Under this model, guide RNA status acts as an instrumental variable (IV) in a two-stage least squares approach to estimate the $D \times D$ matrix of ACEs between genes, $\hat{R}$, where R denotes the true ACE matrix. $\hat{R}$ is derived from the structural equation model (SEM) in Equation 1,

$$\hat{R}_{ij} = \left(\hat{Y}_i^\top \hat{Y}_i\right)^{-1} \hat{Y}_i^\top Y_j. \tag{2}$$

In Equation 2, $\hat{Y}_i$ denotes the predicted gene expression values for gene i from the first-stage regression on guide RNA status, while Y_j denotes the observed gene expression values for gene j. The predicted values capture variation in gene expression explained by the intervention. Since guide RNA status is assumed to be independent of unmeasured confounders, this variation is not associated with these confounders [4]. The predicted values are then used in the second stage to estimate the causal effects between genes.

The authors propose inverse sparse regression (inspre) to estimate $\hat{G}$ from $\hat{R}$. The relationship between G and R, derived from the structural equation model in Equation 1, is

$$G = I - R^{-1} D[1/R^{-1}], \tag{3}$$

where I is the identity matrix, $D[\cdot]$ sets all off-diagonal elements to zero, and the reciprocal is taken element-wise. However, directly applying Equation 3 to $\hat{R}$ is challenging, as $\hat{R}$ may be noisy, ill-conditioned, or noninvertible. Inspre addresses this by estimating two matrices, U and V, where U approximates $\hat{R}$ and V represents its inverse, subject to the constraint $VU = I$. The optimization balances minimizing the difference between U and $\hat{R}$ and obtaining sparse solutions for the inverse matrix V through an L_1 penalty, where the degree of sparsity is controlled by the regularization parameter λ. This optimization problem is solved using the alternating direction method of multipliers [3].

2 Significance

Previous methods for estimating causal effects often rely on strong assumptions that may not be realistic for gene regulatory networks, while also facing computational challenges in high-dimensional settings. For example, some methods assume perfect interventions and the absence of unmeasured confounding [2, 5]. Rather than fixing gene expression at a specific value, the structural equation model in Equation 1 allows for imperfect interventions through the additive term $X\beta$, while gene expression can still depend on other genes through YG [3]. The model also accommodates unmeasured confounding between genes by permitting $\text{Cov}(\gamma_i, \gamma_j) \neq 0$ for $i \neq j$ [3]. Furthermore, some existing methods assume that causal graphs are acyclic, which may not hold for gene regulatory networks containing feedback loops [2, 6, 7]. In contrast, inspre does not require an acyclicity constraint, allowing it to estimate networks with feedback loops [3]. However, an approximate acyclicity constraint can optionally be imposed during estimation. Beyond addressing these limitations, inspre reduces the computational burden by working with the $D \times D$ estimated ACE matrix rather than the full $N \times D$ gene expression matrix during network estimation [2, 6, 7], allowing the method to be applied to larger gene regulatory networks. In simulation studies, Brown et al. [3] found that inspre outperformed other methods in cyclic networks with unmeasured confounding, even under weak interventions. For acyclic networks without confounding, inspre also achieved the highest precision, lowest structural Hamming distance (SHD), and lowest mean absolute error (MAE). Furthermore, inspre required seconds to run, whereas other optimization-based approaches could take up to 10 hours [3]. From a biological perspective, estimating these networks can help researchers understand how causal effects propagate through multiple regulatory pathways [3]. In their application, Brown et al. [3] found that the shortest path between two genes often explained only a small proportion of their total causal effect, highlighting the importance of considering multiple pathways when studying gene regulation.

3 Limitations

The primary assumption underlying inspre is that the true direct-effect network is sparse, meaning that relatively few direct causal relationships exist between genes in a network [8]. Although this assumption is biologically plausible, the authors assume one intervention per gene and uncorrelated interventions for simplicity when deriving their estimator for $(\widehat{R})$. However, a biological limitation in this study is that, in practice, it is often the case where multiple guides are used to target the same gene, demonstrated in their K562 case study [3]. It is unclear how the theoretical results hold in such contexts. To define the edges in their sparse approximation, a threshold of $(v/4)$ is used in the simulations, where (v) denotes the strength of the direct effects between genes. For the K562 case study, a threshold of 0.015 is chosen by taking approximately half of the smallest significant ACE at a 5% false discovery rate [3]. A methodological limitation is that both thresholding procedures appear to be based on judgment rather than a formal statistical criterion. One possible extension could be to develop an IV-aware, decision-theoretic thresholding procedure that accounts for uncertainty in the estimated causal effects and the consequences of incorrectly including or excluding edges. Finally, a technical limitation when using the `inspre` package was an implementation error in its cross-validation procedure, which required modifying the source code to select λ correctly. Although this issue was resolved in our simulations, it could affect users relying on the original package implementation [3].

4 Novel Ideas

One of the primary assumptions underlying inspre is that the direct causal-effect graph is sparse, meaning that most entries of the matrix G are zero [3]. To incorporate sparsity into $\widehat{G}$, inspre uses an L_1 penalty in its optimization objective, which allows the method to estimate the strengths of direct causal effects while shrinking some estimated effects to zero. This approach also avoids directly inverting $\widehat{R}$, which may be noisy or poorly conditioned. The sparsity assumption also reflects the biological expectation that a gene is typically directly regulated by only a small subset of other genes, rather than by every gene in the network [3]. Building on this assumption, the proposed idea is to use the estimated sparse direct causal-effect network ($\widehat{G}$) to reconstruct total causal effects ($\widehat{R}_{\text{sparse}}$), rather than relying solely on the direct IV estimate ($\widehat{R}_{\text{IV}}$). Although IV estimation can be used to estimate total causal effects under appropriate IV assumptions, it does not necessarily guarantee precise estimates. When an instrument is weakly associated with its targeted gene, it may introduce sampling variability and finite-sample bias into the resulting IV estimates, making the estimated total-effect matrix noisy [3]. Furthermore, estimating the total causal effect for each pair of genes separately does not exploit the underlying sparsity of the direct causal-effect network.

Since inspre estimates a sparse network of direct causal effects, it may be possible to use this network to reconstruct total causal effects more accurately and better recover the underlying network structure than through direct IV estimation. Sparsity regularization may reduce estimation noise, although it comes at the cost of potentially shrinking true causal effects toward zero. Figure 1 visualizes this procedure. As a result, the question this qualifying paper seeks to explore is: **can the sparse direct causal-effect network estimated by inspre be used to reconstruct total causal effects more accurately and better recover the underlying network structure than direct IV estimation?**

The research question will be investigated through a proof-of-concept simulation study. The simulation is designed to represent a simple interventional gene perturbation setting. The data were generated according to the model in Equation 1, with $D = 10$ genes connected through a random network constrained to be a directed acyclic graph (DAG). For simplicity, the probability of an edge occurring in the network is set to $p = \frac{1}{9}$ to represent the low-density network. The network edge weights were generated using a PERT distribution, with $v = 0.3$ representing strong direct effects between genes [3]. The coefficient representing the effect of guide RNA interventions was set to $\beta = -2$. The intervention effect was normalized by the standard deviation of gene expression, such that $\beta = -2$ represents a decrease in the mean expression of the targeted gene by two standard deviations [3]. Finally, no latent confounding nodes were included in the simulation ($C = 0$). The number of unperturbed cells was set to 1,000, and for each gene, 200 interventional cells were generated, resulting in a total sample size of 3,000 cells.

To estimate $\widehat{R}_{\text{IV}}$ and $\widehat{G}_\lambda$, the inspre model was fitted to the simulated data using the optional approximate acyclicity constraint, which imposes $D[U] = D[V] = I$. The regularization parameter λ was selected using five-fold cross-validation (CV) by minimizing the structural-model prediction error:

$$L_k(\lambda) = \left\| Y_{(-k)} - Y_{(-k)}\widehat{G}_\lambda^{(k)} - X_{(-k)}\widehat{\beta}_\lambda^{(k)} \right\|_F^2, \tag{4}$$

where $Y_{(-k)}$ and $X_{(-k)}$ represent the held-out observations in fold k, and $\widehat{G}_\lambda^{(k)}$ and $\widehat{\beta}_\lambda^{(k)}$ are estimated using the corresponding training data [3]. Originally, the `predict_inspre()` function in the `inspre` package incorrectly used $\widehat{R}$ instead of $\widehat{G}_\lambda$ when calculating cross-validation prediction errors. This also introduced a dimensional error, because $\widehat{R}$ is a two-dimensional matrix, while

$\widehat{G}_\lambda$ is a three-dimensional array across candidate values of λ. To resolve this, the function was modified to use $\widehat{G}_\lambda$ in the prediction equations. Using Equation 3, $\widehat{R}_{\text{sparse}}$ was reconstructed from the estimated U and V matrices corresponding to the selected λ,

$$\widehat{R}_{\text{sparse}} = D\left[D[\widehat{V}_\lambda]\widehat{U}_\lambda\right]^{-1} D[\widehat{V}_\lambda]\widehat{U}_\lambda. \tag{5}$$

Under the approximate acyclicity constraint, Equation 5 simplifies to $\widehat{R}_{\text{sparse}} = \widehat{U}_\lambda$, since the diagonal entries of $\widehat{U}_\lambda$ and $\widehat{V}_\lambda$ are constrained to one [3]. For the purpose of evaluating estimation accuracy and structural recovery, the diagonal entries of R, $\widehat{R}_{\text{IV}}$, and $\widehat{R}_{\text{sparse}}$ were subsequently set to zero, because the focus is on causal effects between different genes [3].

Estimation accuracy and structural recovery were evaluated using raw and thresholded mean absolute error (MAE) and structural Hamming distance (SHD). Raw MAE measures the average absolute difference between the estimated and true total causal effects without thresholding the edges, while thresholded MAE measures the same difference after thresholding the edges. SHD measures the number of incorrectly identified causal effects, including those with an incorrect sign. Following Brown et al. [3], effects with absolute magnitudes below $v/4$ were treated as zero when calculating thresholded MAE and SHD. The simulation was repeated 100 times, with a new dataset generated in each iteration. Raw MAE, thresholded MAE, and SHD were calculated for both estimators in each iteration, and their mean values across the 100 simulations were used to compare performance. Monte Carlo standard errors (MCSEs) were also calculated to quantify the uncertainty in the estimated mean performance measures.

The estimation accuracy of $\widehat{R}_{\text{sparse}}$ was also evaluated for the subset of true nonzero total causal effects. This was done to determine whether improvements in overall estimation accuracy came at the cost of shrinking true causal effects toward zero. For this subset, raw MAE and mean magnitude error were calculated, where mean magnitude error measures the average signed difference between the estimated and true effect magnitudes.

5 Results

The change in simulated gene expression per unit change in another gene's expression for a single dataset is illustrated in Figure 2. The true R contains relatively few nonzero total causal effects between genes, with some effects being positive and others negative. In contrast, $\widehat{R}_{\text{IV}}$ suggests nonzero effects between nearly all genes. It appears to capture many of the effects in the true R, some effects appear stronger than they actually are, and the estimates remain noisy. The sparse reconstruction, $\widehat{R}_{\text{sparse}}$, produces fewer nonzero effects and more closely resembles the true R.

Across the 100 simulated datasets, $\widehat{R}_{\text{IV}}$ had a higher raw and thresholded MAE than $\widehat{R}_{\text{sparse}}$, indicating that sparse reconstruction provided more accurate estimates of the total causal effects. The mean paired differences in raw and thresholded MAE were 0.01807 (MCSE = 0.00038) and 0.00272 (MCSE = 0.00025), respectively. Since both differences are positive, this supports the idea that sparse reconstruction achieves lower MAE than direct IV estimation. As for SHD, $\widehat{R}_{\text{IV}}$ had a higher value than $\widehat{R}_{\text{sparse}}$, with a mean paired difference of 3.89 (MCSE = 0.242), providing evidence that $\widehat{R}_{\text{sparse}}$ more accurately recovers R.

Although $\widehat{R}_{\text{IV}}$ appears to overestimate the magnitudes of some of the true nonzero total causal effects in the single simulated dataset shown in Figure 2, this pattern is not observed on average across the 100 simulated datasets. Table 2 shows that the mean magnitude error for genuinely nonzero total causal effects was 0.00170 for direct IV estimation, compared with -0.02900 for sparse

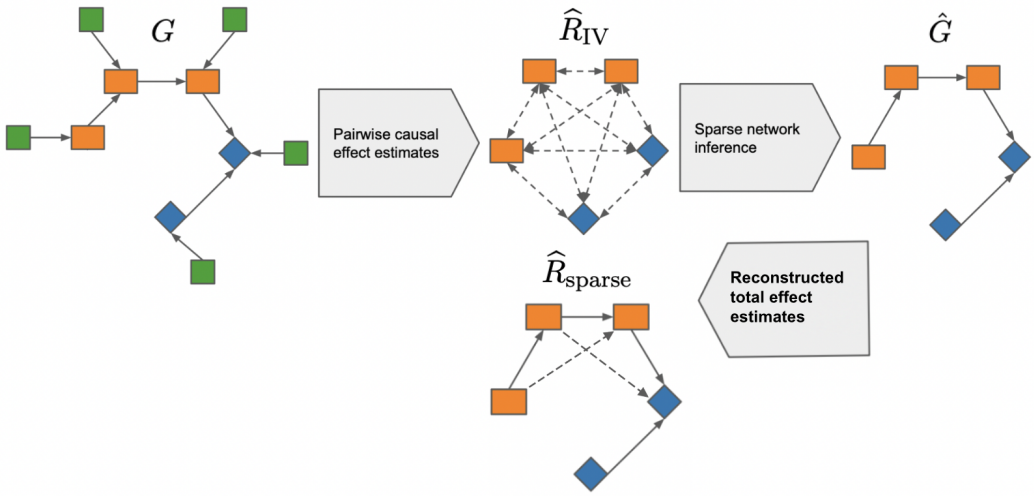


Figure 1: Network G represents the true underlying direct causal-effect network. Interventions (green squares) are used to estimate pairwise total causal effects between genes (orange squares and blue diamonds) using instrumental variables, producing $\widehat{R}_{\text{IV}}$. These estimates are used by inspre to infer a sparse network of direct causal effects, $\widehat{G}$. The proposed approach uses $\widehat{G}$ to reconstruct total causal effects, $\widehat{R}$sparse, including effects arising through indirect causal pathways. Dashed arrows in the reconstructed network represent indirect effects, while solid arrows represent direct effects.

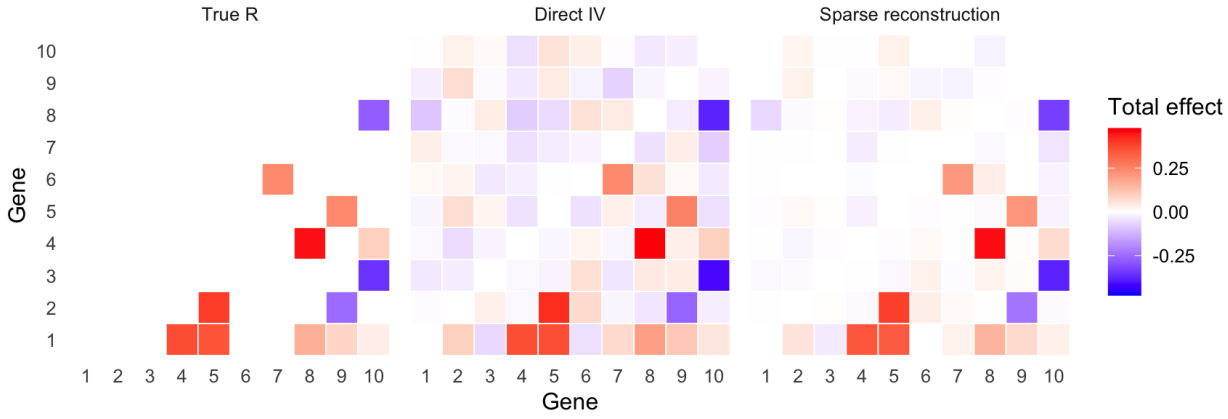


Figure 2: Heatmap of the total causal effects between genes in a single simulated ten-gene network. The first subplot represents the true underlying total effect matrix, R, the second subplot represents the direct IV estimate, $\widehat{R}_{\text{IV}}$, and the third subplot represents the total causal effects reconstructed from the estimated sparse direct-effect network, $\widehat{R}_{\text{sparse}}$. Red indicates positive effects, purple indicates negative effects, and white indicates effects near zero. Total causal effects represent changes in gene expression per unit change in another gene's expression.

Table 1: Raw mean absolute error (MAE), thresholded MAE, structural Hamming distance (SHD), and paired differences between total causal-effect estimators with Monte Carlo standard errors (MCSEs), across 100 simulation repetitions.

Metric	Direct IV $\widehat{R}_{\mathrm{IV}}$	Sparse reconstruction $\widehat{R}_{\mathrm{sparse}}$	Difference
Raw MAE	0.03034 (0.00029)	0.01227 (0.00035)	0.01807 (0.00038)
Thresholded MAE	0.00918 (0.00034)	0.00647 (0.00022)	0.00272 (0.00025)
SHD	5.17 (0.253)	1.28 (0.121)	3.89 (0.242)

reconstruction. Direct IV estimation had a lower raw MAE (0.02891) than sparse reconstruction (0.03561) in this subset as well.

Table 2: Mean magnitude error and raw mean absolute error (MAE) for the subset of true nonzero total causal effects with Monte Carlo standard errors (MCSEs) across 100 simulation repetitions.

Metric	Direct IV $\widehat{R}_{\mathrm{IV}}$	Sparse Reconstruction $\widehat{R}_{\mathrm{sparse}}$
Mean magnitude error	0.00170 (0.00098)	-0.02900 (0.00105)
Raw MAE	0.02891 (0.00066)	0.03561 (0.00077)

6 Discussion

This qualifying paper is focused on investigating if the sparse direct causal-effect network ($\widehat{G}$) estimated using inspre can be used to reconstruct total causal effects with greater estimation accuracy and better structural recovery than direct IV estimation. Results from the simulation study support that for the 10-gene network, the raw MAE and thresholded MAE are smaller for sparse reconstruction compared to the direct IV method, with small MCSEs relative to the differences between the estimators. These findings indicate that sparse reconstruction reduces noise and potentially false-positive edges, making the causal relationships clearer compared to the direct IV method, as visualized in Figure 2. However, when focusing on the subset of true nonzero effects between genes, the raw MAE and the absolute value of the mean magnitude error across the 100 datasets are larger for sparse reconstruction than for direct IV estimation. The mean magnitude error is also negative for sparse reconstruction, suggesting that $\widehat{R}_{\mathrm{sparse}}$ tends to underestimate the magnitudes of the true effects. In terms of structural recovery, the mean paired difference in SHD is 3.89, with a relatively small MCSE of 0.242, consistent with better recovery of the total causal-effect structure through sparse reconstruction. Although sparse reconstruction improves overall estimation accuracy and structural recovery, these improvements come at the cost of reduced accuracy for the true nonzero causal effects.

Whether this trade-off is worthwhile depends on whether the research objective is to recover the overall causal-effect structure or to accurately estimate the magnitudes of total causal effects. For example, in cancer research, TP53 is the most frequently mutated tumour suppressor gene, but directly restoring the function of its protein product is challenging [9]. Thus, reconstructing the total causal effects between genes may help identify upstream genes that could be targeted instead to indirectly affect the protein. Another strategy is synthetic lethality, where disrupting a second gene becomes lethal in a cell already with a genetic defect [9]. For example, MYH10 has been identified as a potential synthetic lethal partner of TP53, so targeting MYH10 may selectively kill TP53-deficient cancer cells [9]. In these applications, identifying potential target genes may be

more important than precisely estimating the magnitudes of individual direct effects, so improved structural recovery could justify this trade-off.

A limitation of this qualifying paper is that the simulation setting is simple and was chosen to favour the performance of inspre. For instance, the chosen network structure is a DAG, even though inspre can handle cyclic networks. The model was also fitted using the optional approximate acyclicity constraint, which the original paper reports can improve convergence behaviour [3]. Both the network effects and intervention strength are strong, because the authors report inspre may miss true edges when both are weak [3]. Additionally, the networks only consisted of 10 genes with no confounders, which does not reflect the complexity of real gene networks. The computational performance and estimation accuracy may change as the numbers of genes increase.

If this topic were developed into a full paper, a natural extension would be to simulate larger, cyclic networks, with weaker network effects, intervention strengths, and unmeasured confounding. Beyond simulation studies, the proposed method could be applied to real CRISPR perturbation data to investigate whether the improvements and trade-offs in estimating the total causal-effect matrix translate to real biological networks. Since the true underlying R is unknown, one option could be to use holdout validation. Here, inspired by the use of held-out data for validation in Yu et al. [10], the data would be split into training and validation sets, ensuring adequate representation of control and intervention cells in each set. In the training set, λ could be selected using cross-validation, and $\widehat{R}_{\text{IV}}$ and $\widehat{R}_{\text{sparse}}$ could then be estimated. In the validation set, an independent $\widehat{R}_{\text{IV}}$ could be estimated and used as a reference for both training-set estimators. The mean squared differences between each training-set estimate and the validation set estimate could then be compared to assess which method agrees more closely with the independent reference. However, since the validation set $\widehat{R}_{\text{IV}}$ itself may have estimation error and potential bias, a smaller mean squared difference does not imply a more accurate estimate of the true R. Nevertheless, this approach could provide an indirect assessment of the relative accuracy of the two estimators.

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